

# [v2.1] Real analysis

## 1. Basics

### Definition 1.1 (Injective)

A function  $f$  is *injective* or *one-to-one* iff:

- $f(a) = f(b) \Rightarrow a = b$ , or equivalently
- $a \neq b \Rightarrow f(a) \neq f(b)$

### Definition 1.2 (Surjective)

A function  $f : A \rightarrow B$  is *surjective* or *onto* iff:

- $f(A) = B$  (its image is its codomain), or equivalently
- for every  $b \in B$  there exists  $a \in A$  s.t.  $f(a) = b$

### Theorem 1.3

Composition of relations preserves the following properties:

- function
- injective
- surjective
- continuous

## 2. Fields and order

### 2.1. Fields

#### Definition 2.1.1 (Field)

A *field*  $\mathbb{F}$  is any set with relations  $+$ ,  $*$ , and  $=$  which satisfies

- If  $x$  and  $y$  are in  $\mathbb{F}$ , so are both  $x + y$  and  $xy$ .
- $x + y = y + x$  and  $xy = yx$ .
- $(x + y) + z = x + (y + z)$  and  $(xy)z = x(yz)$ .
- $x(y + z) = xy + xz$ .
- $w = x$  and  $y + z \Rightarrow w + y = x + z$  and  $wy = xz$ .
- Given  $x, y \in \mathbb{F}$ , there exists  $z \in \mathbb{F}$  s.t.  $x + z = y$  (i.e. all addition problems have solutions).

- There exists some number in  $\mathbb{F}$  which is not equal to 0. If  $x, y \in \mathbb{F}$  s.t.  $x \neq 0$ , then there exists a real number  $z$  s.t.  $xz = y$ . This  $z$  is denoted  $(y/x)$ .
- $=$  is an equivalence relation.

### 2.2. Order

#### Definition 2.2.1 (Order)

A relation  $<$  that satisfies the [total] order axioms:

- **Trichotomy:**  $x < y$ ,  $x = y$ , xor  $x > y$ .
- **Transitivity:** If  $x < y$  and  $y < z$ , then  $x < z$ .

#### Definition 2.2.2 (Dedekind completeness)

An ordered set is *complete*, *Dedekind-complete*, or has the *least upper bound property* iff:

- every subset of the set that has an upper bound has a supremum, or equivalently
- every subset of the set that has a lower bound has an infimum.

### 2.3. Ordered fields

#### Definition 2.3.1 (Ordered field)

A field with an order satisfying

- **Transl.:** If  $x < y$ , then  $x + z < y + z$  for all  $z$ .
- **Pos. closure:** If  $x > 0$  and  $y > 0$ , then  $xy > 0$ .

### 2.4. Real numbers

#### Theorem 2.4.1

There is a unique complete ordered field, called  $\mathbb{R}$ , the *real numbers*.

#### Theorem 2.4.2 (Approximation Property)

Let  $S$  be a nonempty set of reals with a supremum. Then for every real number  $a < \sup S$  there exists  $x \in S$  with  $a < x \leq \sup S$ .

#### Theorem 2.4.3 (Additive Property)

Let  $A$  and  $B$  be sets of reals with suprema. Consider

$$C := \{a + b : a \in A, b \in B\}.$$

Then,

$$\sup C = \sup A + \sup B.$$

#### Theorem 2.4.4 (Comparison Property)

Let  $S$  and  $T$  be two nonempty sets of reals where  $s \leq t$  for all  $s \in S, t \in T$ . Then, if  $T$  has a supremum,  $S$  also has a supremum and  $\sup S \leq \sup T$ .

#### Theorem 2.4.5 (Archimedean property)

The reals are comensural: for any  $x, y \in \mathbb{R}$  with  $x > 0$ , then there is a positive integer  $n$  s.t.  $nx > y$ .

## 2.5. Topology of the reals

### Definition 2.5.1 (Open set)

The following are *open*:

- The empty set
- Open intervals
- Unions of open sets
- Finite intersections of open sets
- The whole space

A set is *closed* if it is the complement of an open set.

The collection of closed sets is closed under finite union and arbitrary intersection.

### Definition 2.5.2 (Continuity)

Let  $f : X \rightarrow Y$  be a function from topological space  $X$  to topological space  $Y$ . Then,  $f$  is continuous iff  $f^{-1}(A) \subset X \Leftrightarrow A$  is open in  $Y$ .

## 2.6. Absolute values

### Theorem 2.6.1 (Corollaries of the Triangle Inequality)

For any  $x, y \in \mathbb{R}$ :

$$\begin{aligned}|x + y| &\leq |x| + |y| \\ |x - y| &\leq |x| + |y| \\ ||x| + |y|| &\leq |x - y|\end{aligned}$$

## 3. Sequences

### Definition 3.1 (Limit of a sequence)

Let  $\{x_n\}$  be a sequence.  $L$  is the **limit**, denoted  $\lim_{n \rightarrow \infty} x_n$ , of the sequence iff for every  $\varepsilon > 0$  there exists  $N$  s.t. whenever  $n > N$ ,  $|x_n - L| < \varepsilon$ .

Note - monotonic does not require strictly increasing/decreasing, it can be stable at some points.

### Theorem 3.2

A bounded monotonically increasing sequence converges to its supremum.

A bounded monotonically decreasing sequence converges to its infimum.

### Theorem 3.3

Every nonempty bounded subset of the reals has an increasing sequence converging to its supremum and a decreasing sequence converging to its infimum.

## 3.1. Subsequences

### Theorem 3.1.1

The following are equivalent:

1. The sequence  $\{x_n\}$  converges.
2. The  $t$ -tail  $\{x_{n+t}\}$  converges for every  $t$ .
3. There is some  $t$  for which the  $t$ -tail converges.

## 3.2. Limit properties and theorems

### Theorem 3.2.1

Let  $\{x_n\} \rightarrow x$  and  $\{y_n\} \rightarrow y$ , s.t.  $x_n \leq y_n$  for all  $n$ . Then,  $x \leq y$ .

### Theorem 3.2.2 (Squeeze Theorem)

Let  $\{a_n\}$ ,  $\{b_n\}$ , and  $\{c_n\}$  be sequences s.t.  $a_n \leq b_n \leq c_n$  for all  $n$  and  $a_n \rightarrow L$  and  $c_n \rightarrow L$ . Then  $b_n \rightarrow L$ .

### Theorem 3.2.3

Let  $\{x_n\} \rightarrow x$  and  $\{y_n\} \rightarrow y$ . Then

1.  $\{x_n + y_n\} \rightarrow x + y$ .
2.  $\{x_n - y_n\} \rightarrow x - y$ .
3.  $\{x_n y_n\} \rightarrow xy$ .
4.  $\{x_n / y_n\} \rightarrow x / y$  if  $y_n \neq 0$  and  $y \neq 0$  for all  $n$ .

### Theorem 3.2.4

Let  $\{x_n\}$  be a sequence and  $x$  a number. Then,

$$\{x_n\} \rightarrow x \iff \{x_n - x\} \rightarrow 0.$$

### Theorem 3.2.5 (Convergence of geometric series)

Let  $c > 0$ . Then,

1.  $c < 1 \implies \lim_{n \rightarrow \infty} c^n = 0$ .
2.  $c > 1 \implies \{c^n\}$  is unbounded.

### Theorem 3.2.6 (Ratio Test)

Let  $\{x_n\}$  be an always-nonzero sequence s.t.

$$\lim_{n \rightarrow \infty} \frac{|x_{n+1}|}{|x_n|} = L.$$

Then,

- $L < 1 \implies \{x_n\} \rightarrow 0$ .
- $L > 1 \implies \{x_n\}$  is unbounded.

### Theorem 3.2.7

$$\lim_{n \rightarrow \infty} n^{\frac{1}{n}} = 1$$

## 3.3. Limsup and liminf

### Definition 3.3.1 (Limsup and liminf)

Let  $\{x_n\}$  be a bounded sequence. For each  $n$ , define

$$\begin{aligned}a_n &:= \sup\{x_k : k \geq n\} \\ b_n &:= \inf\{x_k : k \geq n\}\end{aligned}$$

Thus,  $a_n$  is above the  $n$ -tail of the sequence, while  $b_n$  is below it.

Define

$$\begin{aligned}\limsup_{n \rightarrow \infty} x_n &= \lim_{n \rightarrow \infty} a_n \\ \liminf_{n \rightarrow \infty} x_n &= \lim_{n \rightarrow \infty} b_n.\end{aligned}$$

We know that  $\limsup x_n \geq \liminf x_n$ .

### Theorem 3.3.2

If  $\{x_n\}$  is bounded, then:

- there is a subsequence  $\{x_{n_k}\} \rightarrow \limsup x_n$ .
- there is a subsequence  $\{x_{m_k}\} \rightarrow \liminf x_n$ .

### Theorem 3.3.3

Let  $\{x_n\}$  be a bounded sequence with  $\limsup x_n = \liminf x_n = L$ . Then,  $\{x_n\} \rightarrow L$ .

### Theorem 3.3.4

$\{x_n\} \rightarrow L$  iff every subsequence converges to  $L$ .

## 3.4. Bolzano-Weierstrass

### Theorem 3.4.1 (Bolzano-Weierstrass for Sequences)

Every bounded sequence of real numbers (or in  $\mathbb{R}^n$ ) has a convergent subsequence.

In topology terms, a subset of  $\mathbb{R}^n$  is sequentially compact iff it is closed and bounded.

### Definition 3.4.2 (Open ball)

Let  $x \in \mathbb{R}^n$ . The ball  $B(x, \varepsilon)$  of radius  $\varepsilon$  about  $x$  is defined as

$$B(x, \varepsilon) := \{y \in \mathbb{R}^n : |x - y| < \varepsilon\}.$$

### Theorem 3.4.3 (Bolzano-Weierstrass theorem)

Every bounded infinite subset of  $\mathbb{R}^n$  has an accumulation point.

### Theorem 3.4.4 (Cantor Intersection Theorem)

Let  $S_1, S_2, \dots$  be a sequence of bounded closed subsets of  $\mathbb{R}^n$  nested as follows:  $S_1 \supset S_2 \supset \dots$ . Then, the intersection  $S_1 \cap S_2 \cap \dots$  is closed and nonempty.

## 3.5. Cauchy sequences

### Definition 3.5.1 (Cauchy sequence)

A sequence  $\{x_n\}$  is called **Cauchy sequence** iff for every  $\varepsilon > 0$ , there exists  $N$  s.t. for all  $m, n > N$ ,  $|x_n - x_m| < \varepsilon$ .

In short, beyond a certain term, all the following terms are arbitrarily close.

Every Cauchy sequence is bounded. A sequence of reals converges iff it is a Cauchy sequence.

## 4. Series

### 4.1. Basic convergence tests

#### Theorem 4.1.1 ( $n^{\text{th}}$ term test for divergence)

If  $\sum x_n$  converges, then  $x_n \rightarrow 0$ .

Equivalently, if  $x_n \not\rightarrow 0$ ,  $\sum x_n$  diverges.

### Theorem 4.1.2 (Linearity of series)

Let  $\sum x_n = x$  and  $\sum y_n = y$ . Then,  $\sum (x_n + y_n) = x + y$ .

For any  $c \in \mathbb{R}$ ,  $\sum (cx_n) = cx$ .

### Theorem 4.1.3 (p-series)

$\sum_{n=1}^{\infty} \frac{1}{n^p}$  converges iff  $p > 1$ .

## 4.2. Absolute convergence

### Lemma 4.2.1

If  $x_n \geq 0$  for all  $n$ , then  $\sum x_n$  converges iff the sequence of partial sums is bounded above. Furthermore, each partial sum is less than or equal to the sum.

### Definition 4.2.2 (Absolute convergence)

Iff  $\sum |x_n|$  converges, then  $\sum x_n$  **absolutely converges**. If  $\sum |x_n|$  diverges, but  $\sum x_n$  converges, then  $\sum x_n$  **conditionally converges**.

### Theorem 4.2.3 (Comparison test for positive series)

Let  $\sum x_n$  and  $\sum y_n$  be series with  $0 \leq x_n \leq y_n$  for all  $n$ . Then:

1. If  $\sum y_n$  converges, then  $\sum x_n \leq \sum y_n$  ( $\sum x_n$  converges).
2. If  $\sum x_n$  diverges, so does  $\sum y_n$ .

## 4.3. Ratio test and root test

### Theorem 4.3.1 (Ratio test)

Let  $\sum x_n$  be a series of nonzero terms. Suppose that

$$L := \lim_{n \rightarrow \infty} \frac{|x_{n+1}|}{|x_n|}$$

exists. Then,

- If  $L < 1$ ,  $\sum x_n$  converges absolutely.
- If  $L > 1$ ,  $\sum x_n$  diverges.
- If  $L = 1$ , the test gives no information.

### Theorem 4.3.2 (Root test)

Let  $\sum x_n$  be a series and  $L := \limsup \sqrt[n]{|x_n|}$  (which may be infinite). Then,

- If  $L < 1$ ,  $\sum x_n$  converges absolutely.
- If  $L > 1$ ,  $\sum x_n$  diverges.
- If  $L = 1$ , the test gives no information.

## 4.4. Alternating series

### Theorem 4.4.1 (Alternating series test)

Let  $x_n \rightarrow 0$  be monotone decreasing. Then,  $\sum (-1)^n x_n$  converges.

### Theorem 4.4.2

Let  $\sum x_n$  be an absolutely convergent series converging to  $L$ . Let  $\sigma : \mathbb{N} \rightarrow \mathbb{N}$  be a bijection. Then,  $\sum x_{\sigma n}$  is also absolutely convergent and converges to  $L$ .

In short, if the series is absolutely convergent, we can rearrange the terms and the sum is the same.

### Theorem 4.4.3 (Alternating Series Rearrangement Theorem)

Let  $\sum x_n$  be a series that satisfies the alternating series test but which is not absolutely convergent. Then, for any  $L \in \mathbb{R}$  there is a rearrangement of the series that sums to  $L$ .

## 4.5. Multiplying series

### Definition 4.5.1 (Cauchy product)

The **Cauchy product** of  $\sum a_n$  and  $\sum b_n$  is defined as

$$\sum_{k=1}^n a_n b_{n+1-k}.$$

Essentially, we collect the terms of the product of  $\sum a_n$  with  $\sum b_n$  in diagonal strips when the terms are arranged in a grid.

### Theorem 4.5.2 (Mertens' theorem)

Let  $\sum a_n \rightarrow A$  and  $\sum b_n \rightarrow B$ . Then if at least one of these is absolutely convergent, then  $\sum c_n \rightarrow AB$ , where  $\sum c_n$  is the Cauchy product of  $\sum a_n$  and  $\sum b_n$ .

Additionally, if both converge absolutely, so does the sum of  $c_n$ s.

## 4.6. Power series

### Definition 4.6.1 (Power series)

Given  $x_0 \in \mathbb{R}$  and a sequence  $a_n$ , we can create the **power series**, a function of  $x$ :

$$\sum_{n=0}^{\infty} a_n (x - x_0)^n.$$

A power series **converges** iff there exists  $x \neq x_0$  that makes the series converge. If  $x = x_0$  makes the series converge, then the series converges for any  $x$ .

### Theorem 4.6.2

A power series either diverges for all  $x \neq x_0$ , converges absolutely for all  $x$ , or there exists a real number  $\rho$ , called the **radius of convergence**, s.t. the series converges absolutely on  $(x_0 - \rho, x_0 + \rho)$  and diverges if  $|x - x_0| > \rho$ .

When  $x = x_0 + \rho$  or  $x = x_0 - \rho$ , this theorem is inconclusive.

# 5. Limits and continuity

## 5.1. Limits

### Definition 5.1.1 (Accumulation point)

$x$  is a **accumulation point** of  $S$  iff

- every ball  $B(x, \varepsilon)$  contains a point of  $S$  other than  $x$ , or, equivalently,
- there is a sequence of points in  $S - \{x\}$  converging to  $x$ .

### Definition 5.1.2 (Limit)

Let  $S$  be a subset of  $\mathbb{R}$  and  $c$  be an accumulation point of  $S$ . Let  $f : S \rightarrow \mathbb{R}$  be a function.

The **limit** of  $f(x)$  as  $x \rightarrow c$  is  $L$  iff for each  $\varepsilon > 0$ , there exists a  $\delta > 0$  s.t. whenever  $x \in S - \{c\}$  and  $|x - c| < \delta$ , then  $|f(x) - L| < \varepsilon$ .

### Theorem 5.1.3

Let  $S$  be a subset of  $\mathbb{R}$  and  $c$  be an accumulation point of  $S$ . Let  $f : S \rightarrow \mathbb{R}$  be a function.

Then  $f(x) \rightarrow L$  as  $x \rightarrow c$  iff for every sequence  $\{x_n\}$  in  $S - \{c\}$  whose limit is  $c$ , the sequence  $\{f(x_n)\} \rightarrow L$ .

## 5.2. Continuity

### Definition 5.2.1 (Continuity)

Let  $S \subset \mathbb{R}$ ,  $c \in S$ , and  $f : S \rightarrow \mathbb{R}$  be a function.

Then,  $f$  is **continuous** at  $c$  iff for every  $\varepsilon > 0$ , there is a  $\delta > 0$  s.t. whenever  $|x - c| < \delta$ , then  $|f(x) - f(c)| < \varepsilon$ .

### Theorem 5.2.2

Let  $S \subset \mathbb{R}$  and  $f : S \rightarrow \mathbb{R}$  be a function. Let  $c$  be a point in  $S$ . Then:

1. If  $c$  is not an accumulation point of  $S$ , then  $f$  is continuous at  $c$ .

2. If  $c$  is an accumulation point of  $S$ , then  $f$  is continuous at  $c$  iff  $\lim_{x \rightarrow c} f(x) = f(c)$ .
3.  $f$  is continuous at  $c$  iff for every sequence  $\{x_n\} \in S$  with limit  $c$ ,  $f(x_n) \rightarrow f(c)$ .

## 5.3. Big Continuity Theorems

### Definition 5.3.1 (Bounded function)

A function  $f : S \rightarrow \mathbb{R}$  is bounded iff its image is bounded.

### Lemma 5.3.2

Let  $[a, b]$  be a closed finite interval, and  $f$  continuous on  $[a, b]$ . Then  $f$  is bounded.

### Theorem 5.3.3 (Extreme Value Theorem)

Let  $f(x)$  be a continuous function defined on a closed interval  $[a, b]$ . Then,  $f$  achieves an absolute maximum and minimum on  $[a, b]$ .

### Lemma 5.3.4

Let  $f(x)$  be continuous on the closed bounded interval  $[a, b]$ . If  $f(a) < 0$  and  $f(b) > 0$ , then there exists  $c \in [a, b]$  s.t.  $f(c) = 0$ .

### Theorem 5.3.5 (Intermediate Value Theorem)

Let  $f$  be continuous on closed, bounded  $[a, b]$ . Let  $y \in \mathbb{R}$  s.t.  $f(a) < y < f(b)$  or  $f(a) > y > f(b)$ . Then, there exists  $c \in (a, b)$  with  $f(c) = y$ .

## 5.4. Uniform continuity

### Definition 5.4.1 (Uniform continuity)

Let  $S \subset \mathbb{R}$  and  $f : S \rightarrow \mathbb{R}$  be a function. For any  $\varepsilon > 0$ ,  $f$  is **uniformly continuous** on  $S$  iff we can find  $\delta$  s.t. for all  $x, c \in S$  with  $|x - c| < \delta$ , then  $|f(x) - f(c)| < \varepsilon$ .

This differs from continuity iff the same  $\delta$  works for every point in  $S$ .

Uniform continuity obviously implies continuity.

### Theorem 5.4.2

If  $f : [a, b] \rightarrow \mathbb{R}$  is a continuous function defined on a closed bounded interval, it is uniformly continuous.

### Lemma 5.4.3

If  $f : S \rightarrow \mathbb{R}$  is uniformly continuous and  $\{x_n\}$  is a Cauchy sequence in  $S$ , then  $\{f(x_n)\}$  is a Cauchy sequence.

### Theorem 5.4.4

Let  $f : (a, b) \rightarrow \mathbb{R}$ . Then  $f$  is uniformly continuous iff it has a right-handed limit at  $a$ , a left-handed limit at  $b$ , and the extension of  $f$  to the closed interval  $[a, b]$  is continuous.

## 5.5. Lipschitz continuity

### Definition 5.5.1 (Lipschitz continuity)

Let  $f : S \rightarrow \mathbb{R}$  be a function. If there exists  $K \in \mathbb{R}$  s.t. for every  $x, y \in S$ ,  $|f(x) - f(y)| < K|x - y|$  then  $f$  is **Lipschitz continuous**.

### Theorem 5.5.2

Lipschitz continuity implies uniform continuity.

## 6. Derivatives

### Definition 6.1 (Derivative)

Let  $f(x)$  be a real-valued function defined on an open interval  $(a, b)$ , and let  $x_0 \in (a, b)$ . The **derivative** of  $f$  at  $x_0$  is

$$\frac{df}{dx}(x_0) := f'(x_0) := \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}.$$

Iff this limit is defined, the function is **differentiable** at  $x_0$ .

### Theorem 6.2 (Continuity theorem)

Let  $f$  be defined on  $(a, b)$  with  $x_0 \in (a, b)$ . Then,  $f$  is differentiable at  $x_0$  iff there is a function  $f^*$  (which depends on  $f$  and on  $x_0$ ) s.t.

$$f(x) = f(x_0) + f^*(x)(x - x_0)$$

with  $f^*$  defined on  $(a, b)$  and continuous at  $x_0$ , with  $f^*(x_0) = f'(x_0)$ .

If  $f$  is continuous,  $f^*$  can be defined as

$$f^*(x) := \begin{cases} \frac{f(x) - f(x_0)}{x - x_0} & \text{if } x \neq x_0 \\ f'(x_0) & \text{if } x = x_0. \end{cases}$$

### Theorem 6.3 (Sum, product, and quotient rules)

Let  $f$  and  $g$  be functions defined on an interval  $(a, b)$ , both differentiable at the point  $x_0 \in (a, b)$ . (In particular, the derivatives there cannot be infinite.) Then, the following derivatives exist and take on the following values:

- $(f \pm g)'(x_0) = f'(x_0) \pm g'(x_0)$ .
- $(f * g)'(x_0) = f(x_0)g'(x_0) + f'(x_0)g(x_0)$ .
- $(f/g)'(x_0) = \frac{g(x_0)f'(x_0) - f(x_0)g'(x_0)}{g(x_0)^2}$ .

### Theorem 6.4 (Chain rule)

Let  $f : (a, b) \rightarrow \mathbb{R}$  and  $g : f(a, b) \rightarrow \mathbb{R}$ . Let  $x_0 \in (a, b)$  with  $f$  differentiable at  $x_0$  and  $f(x_0)$  contained in the interior of  $f(a, b)$ , and let  $g$  be differentiable at  $f(x_0)$ . Then,  $(g \circ f)(x)$  is differentiable at  $x_0$ , and

$$(g \circ f)'(x_0) = g'(f(x_0))f'(x_0),$$

or equivalently,

$$\left. \frac{dg}{dx} \right|_{x_0} = \left. \frac{dg}{df} \right|_{f(x_0)} * \left. \frac{df}{dx} \right|_{x_0}.$$

## 6.1. Mean value theorem

### Theorem 6.1.1

Let  $f'(c) > 0$  and  $f$  continuous in some interval about  $c$ . Then, there is a  $\delta > 0$  s.t.:

- $c < x < c + \delta \implies f(c) < f(x)$
- $c - \delta < x - c \implies f(x) < f(c)$

Equivalently, if the derivative is positive, then the function is increasing at that point.

Similarly, if the derivative is negative, then the function is decreasing at that point.

### Corollary 6.1.1.1

If  $f(c)$  is a local min or local max, then  $f'(c) = 0$ .

### Definition 6.1.2 (Critical point)

A point where the derivative of the function is zero.

### Theorem 6.1.3 (Rolle's theorem)

Let  $f(x)$  be continuous on  $[a, b]$  and differentiable on  $(a, b)$ , s.t.  $f(a) = f(b)$ . Then, there exists  $c \in (a, b)$  for which  $f'(c) = 0$ .

### Theorem 6.1.4 (Mean value theorem)

Let  $f$  be continuous on  $[a, b]$  and differentiable on  $(a, b)$ . Then, there exists some point  $c \in (a, b)$  for which

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

### Theorem 6.1.5 (Cauchy's mean value theorem)

Let  $f(t)$  and  $g(t)$  be two functions continuous on  $[a, b]$  and differentiable on  $(a, b)$ . Then, there exists some point  $c \in (a, b)$  for which

$$f'(c)(g(b) - g(a)) = g'(c)(f(b) - f(a)).$$

### Theorem 6.1.6

Let  $f(x)$  be continuous on  $[a, b]$  and differentiable on  $(a, b)$ . Then:

- $f'(x) > 0$  for all  $x \in (a, b) \iff f$  is strictly increasing
- $f'(x) < 0$  for all  $x \in (a, b) \iff f$  is strictly decreasing
- $f'(x) = 0$  for all  $x \in (a, b) \iff f$  is constant

Let  $g(x)$  also be continuous on  $[a, b]$  and differentiable on  $(a, b)$ . Then:

- $g'(x) = f'(x)$  for all  $x \in (a, b) \iff f(x) - g(x)$  is constant

### Theorem 6.1.7

Let  $f(x)$  be continuous on  $(a, b]$  and differentiable on  $(a, b)$ , with  $\lim_{x \rightarrow b^-} f'(x) = L$ . Then,  $f$  is left-hand differentiable at  $b$ , and  $f'_-(b) = L$ .

A similar theorem exists for right-hand derivatives.

### Theorem 6.1.8 (Darboux theorem)

Let  $f(x)$  be a function that is differentiable on  $[a, b]$ . Let  $y$  be a value s.t.  $f'(a) < y < f'(b)$  or  $f'(a) > y > f'(b)$ . Then there is some  $c \in (a, b)$  s.t.  $f'(c) = y$ .

### Corollary 6.1.8.1

- $f'$  cannot change signs without going through zero.
- If  $f'(x) \neq 0$  for any  $x$ , then  $f$  is monotonic.

- If  $f'$  itself is monotonic then it must be continuous:
  - the only kinds of discontinuities of monotonic functions are jumps
  - derivatives can't make jumps

## 6.2. Taylor series

### Theorem 6.2.1 (Taylor's formula with the Lagrange form of the remainder)

Let  $f : (a, b) \rightarrow \mathbb{R}$  have  $n + 1$  derivatives on the interval  $(a, b)$ , and let  $x_0$  be a point in  $(a, b)$ . Then, for any point  $y$  in  $(a, b)$  **Taylor's formula** states that

$$f(y) = p_n(y) + R_n(y; x_0).$$

We define the  **$n$ th Taylor polynomial centered at  $x_0$**  as:

$$p_n(x) := \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k.$$

The **Taylor series** is the limit of the Taylor polynomials.

We define the **Lagrange form of the remainder** as:

$$R_{n(x; x_0)} := \frac{f^{(n+1)}(c)(y - x_0)^{n+1}}{(n + 1)!}$$

for some  $c$  between  $y$  and  $x_0$  [this theorem does not specify which  $c$  this is].

## 7. Integrals

### 7.1. Partitions

#### Definition 7.1.1 (Partition)

A **partition**  $P$  of  $[a, b]$  is a finite set of points,  $\{x_0, x_1, \dots, x_n\}$  with  $a = x_0 < x_1 < \dots < x_n = b$ .

The  $k$ th interval of the partition is  $[x_{k-1}, x_k]$ , with width  $\Delta x_k := x_k - x_{k-1}$ .

#### Definition 7.1.2 (Norm/mesh)

The **norm** or **mesh** of a partition is the width of its largest interval, denoted

$$\|P\| := \max\{\Delta x_k : k \leq n\}.$$

#### Definition 7.1.3 (Refinement)

$\tilde{P}$  is a **refinement** of  $P$  iff  $P \subseteq \tilde{P}$ .

## 7.2. Darboux integral

### Definition 7.2.1 (Darboux sum)

Let  $f : [a, b] \rightarrow \mathbb{R}$  be bounded and  $P$  a partition of  $[a, b]$ . Then, define

$$m_k := \inf\{f(x) : x_{k-1} \leq x \leq x_k\}$$

$$M_k := \sup\{f(x) : x_{k-1} \leq x \leq x_k\}.$$

Define the **upper** and **lower Darboux sums**, respectively, as

$$L(P, f) := \sum_{k=1}^n m_k \Delta x_k$$

$$U(P, f) := \sum_{k=1}^n M_k \Delta x_k.$$

### Definition 7.2.2 (Upper and lower Darboux integrals)

Let  $f : [a, b] \rightarrow \mathbb{R}$  be bounded. Then, define the **upper** and **lower Darboux integrals** as follows:

$$\int_a^b f(x) dx := \inf\{U(P, f) : P \text{ is a partition of } [a, b]\}$$

$$\int_a^b f(x) dx := \sup\{L(P, f) : P \text{ is a partition of } [a, b]\}.$$

### Definition 7.2.3 (Darboux integrability)

If  $f : [a, b] \rightarrow \mathbb{R}$  is bounded, and  $\int_a^b f(x)dx = \overline{\int_a^b f(x)dx}$ , then we say  $f$  is **Darboux-integrable** ( $f \in \mathcal{R}[a, b]$ ), and we define  $\int_a^b f(x)dx$  to be the common value of the lower and upper Darboux integral.

## 7.3. Riemann integral

### Definition 7.3.1 (Riemann sum)

Let  $f : [a, b] \rightarrow \mathbb{R}$  be bounded and  $P$  a partition of  $[a, b]$ .

For each interval  $[x_{i-1}, x_i]$ , select any  $t_i$  in that interval. Then, the **Riemann sum** of  $f$  under this selection of  $t_i$  is

$$\sum_{i=1}^n f(t_i) \Delta x_i.$$

### Definition 7.3.2 (Riemann integrability)

$f$  is **Riemann-integrable** ( $f \in \mathcal{R}[a, b]$ ) iff there exists some number  $I$  s.t. for every  $\varepsilon > 0$ , there exists a partition  $P_0$  s.t. for every refinement  $P$  of  $P_0$  and every choice of  $t_i$  on  $P$ , the corresponding Riemann sum is within  $\varepsilon$  of  $I$ .

### Theorem 7.3.3

Riemann integrability is equivalent to Darboux integrability.

## 7.4. Integrability

### Theorem 7.4.1

Let  $f : [a, b] \rightarrow \mathbb{R}$  be bounded. Then,  $f$  is Darboux-integrable iff for every  $\varepsilon > 0$ , there exists a partition  $P$  s.t.  $U(P, f) - L(P, f) < \varepsilon$ .

### Theorem 7.4.2

Let  $f : [a, b] \rightarrow \mathbb{R}$  be bounded and monotonic. Then,  $f \in \mathcal{R}[a, b]$ .

## 7.5. Core properties of the integral

### Theorem 7.5.1 (Additivity)

Let  $a < b < c$  and  $f : [a, c] \rightarrow \mathbb{R}$  be a bounded function. Then,

$$f \in \mathcal{R}[a, c] \iff f \in \mathcal{R}[a, b] \text{ and } f \in \mathcal{R}[b, c],$$

and if  $f$  is integrable on the whole interval then

$$\int_a^c f = \int_a^b f + \int_b^c f.$$

### Theorem 7.5.2 (Linearity)

Let  $f, g \in \mathcal{R}[a, b]$ , and  $\alpha$  any real constant. Then,

$$\int_a^b \alpha f(x) + g(x)dx = \alpha \int_a^b f(x)dx + \int_a^b g(x)dx.$$

### Theorem 7.5.3 (Monotonicity)

Let  $f$  and  $g$  be bounded functions on  $[a, b]$ . Assume  $f(x) \leq g(x)$  for all  $x \in [a, b]$ . Then,  $\int_a^b f < \int_a^b g$  and  $\overline{\int_a^b f} < \overline{\int_a^b g}$ . If both are in  $\mathcal{R}[a, b]$  then  $\int_a^b f < \int_a^b g$ .

## 7.6. Additional properties of the integral

### Theorem 7.6.1

Let  $f \in [a, b]$ . Let  $f(x) \in [m, M]$  for all  $x \in [a, b]$ . Then,

$$m(b-a) \leq \int_a^b f(x)dx \leq M(b-a).$$

### Theorem 7.6.2 (Integral of a constant)

$$\int_a^b cdx = c(b-a).$$

### Theorem 7.6.3 (Integral of a step function)

Let

$$f(x) = \begin{cases} r & \text{for } x < c \\ s & \text{for } x = c, \\ t & \text{for } x > c \end{cases}$$

with  $c \in [a, b]$  and  $f : [a, b] \rightarrow \mathbb{R}$ .

Then,

$$\int_a^b f(x)dx = r(c-a) + t(b-c).$$

### Definition 7.6.4

$$\int_a^b f(x)dx := - \int_b^a f(x)dx.$$

## 7.7. Integrals of continuous functions

### Lemma 7.7.1

A bounded function that has only finitely many discontinuities is integrable.

### Lemma 7.7.2

Let  $f : [a, b] \rightarrow \mathbb{R}$  be bounded. Let  $\{a_n\} \rightarrow a$  and  $\{b_n \rightarrow b\}$  be sequences in  $[a, b]$ , with  $a_n < b_n$  for all  $n$ . Then, if  $f \in \mathcal{R}[a_n, b_n]$  for all  $n$ , then  $f \in \mathcal{R}[a, b]$  and  $\int_a^b f = \lim_{n \rightarrow \infty} \int_{a_n}^{b_n} f$ .

Essentially, bounded functions need only be integrable inside the interval to be integrable on the entire interval.

### Theorem 7.7.3

Continuous functions are integrable.

## 7.8. Fundamental theorem of calculus

### Theorem 7.8.1 (First Fundamental Theorem of Calculus)

Let  $F : [a, b] \rightarrow \mathbb{R}$  be continuous and differentiable\* on  $(a, b)$ , with  $F' \in \mathcal{R}[a, b]$ . Then,

$$\int_a^b F'(x)dx = F(b) - F(a).$$

\* $F$  may have finitely many points where it is not differentiable. It must be continuous over the whole interval.

### Theorem 7.8.2 (Second Fundamental Theorem of Calculus)

Let  $f \in \mathcal{R}[a, b]$ . Define  $F(x) := \int_a^x f$ . Then,  $F$  is continuous on  $[a, b]$ , and if  $f$  is continuous at  $c$ , then  $F$  is differentiable at  $c$  and  $F'(c) = f(c)$ .

### Theorem 7.8.3 (u-substitution)

Let  $f : [c, d] \rightarrow \mathbb{R}$  be continuous and  $g : [a, b] \subset [c, d]$  continuously differentiable. Then,

$$\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du.$$

Note that in this case,  $u = g(x)$  and  $du = g'(x)dx$ .

## 8. Sequences of functions

### 8.1. Pointwise convergence

#### Definition 8.1.1 (Pointwise convergence)

Let  $\{f_1, f_2, \dots\}$  be a sequence of functions. Let  $S$  be the set of all points  $x$  for which the sequence

$\{f_1(x), f_2(x), \dots\}$  converges. Then,  $S$  is the domain of the limit function

$$f(x) := \lim_{n \rightarrow \infty} f_n(x).$$

$\{f_n\}$  **converges pointwise** to  $f$  on  $S$ .

### 8.2. Uniform convergence

#### Definition 8.2.1 (Uniform convergence)

Let  $\{f_n\}$  be a sequence of functions.  $\{f_n\}$  **converges uniformly** to  $f$  on  $S$  iff for every  $\varepsilon > 0$  there exists  $N$  s.t.  $n > N$  implies  $|f(x) - f_n(x)| < \varepsilon$  for all  $x \in S$ .

#### Definition 8.2.2

Let  $\{f_n\}$  be a sequence of functions on  $S$ . Let  $M > 0$  be a constant so that  $|f_{n(x)}| \leq M$  for all  $x$  and  $n$ . Then,  $\{f_n\}$  is **uniformly bounded** by  $M$  by  $S$ .

#### Theorem 8.2.3

Let  $f_n \rightarrow f$  uniformly on  $S$  and let each  $f_n$  be bounded. Then,  $f$  is bounded and the sequence  $\{f_n\}$  is uniformly bounded on  $S$ .

#### Definition 8.2.4 (Uniform norm)

Let  $f$  be a bounded function on domain  $S$ . Then, define the **uniform norm** or **infinity norm** of  $f$  on  $S$  is

$$\|f\|_\infty = \|f\|_u := \sup\{|f(x)| : x \in S\}.$$

#### Theorem 8.2.5

A sequence of bounded functions  $f_n$  converges uniformly to  $f$  iff

$$\lim_{n \rightarrow \infty} \|f_n - f\|_\infty = 0.$$

### Theorem 8.2.6

Let  $\{f_n\}$  be a sequence of functions. Then there exists a function  $f$  s.t.  $f_n \rightarrow f$  uniformly iff the sequence satisfies the **Cauchy condition**: for every  $\varepsilon > 0$  there exists  $N$  s.t. if  $m, n > N$ , then  $|f_n(x) - f_m(x)| < \varepsilon$  for all  $x$ .

#### Definition 8.2.7

A series converges uniformly iff its sequence of partial sums converges uniformly.

### Theorem 8.2.8 (Weierstraß M-test)

Let  $M_n$  be a sequence of nonnegative numbers s.t.  $|f_n(x)| \leq M_n$  for all  $x$  and  $n$ . Then,  $\sum f_n$  converges uniformly if  $\sum M$  converges.

## 8.3. Continuity, integrability, and differentiability of limit functions

### Theorem 8.3.1 (Interchanging limits)

When  $f_n \rightarrow f$  uniformly, then

$$\begin{aligned} \lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} \lim_{n \rightarrow \infty} f_n(x) \\ &= \lim_{n \rightarrow \infty} \lim_{x \rightarrow a} f_n(x) = \lim_{n \rightarrow \infty} f_n(a) = f(a). \end{aligned}$$

### Theorem 8.3.2

Let  $\{f_n\}$  be a sequence of continuous functions  $S \rightarrow \mathbb{R}$ , where  $S$  is some domain. If the sequence converges uniformly to a function  $f$ , then  $f$  is also continuous.

### Theorem 8.3.3

Let  $\{f_n\}$  be a sequence of integrable functions on  $[a, b]$  that converges uniformly to  $f$ . Then  $f$  is integrable, and

$$\int_a^b f = \lim_{n \rightarrow \infty} \int_a^b f_n.$$

### Theorem 8.3.4

Let each term of  $\{f_n\}$  be differentiable at each point of  $(a, b)$  and assume that these derivatives converge  $f'_n \rightarrow g$  uniformly on  $(a, b)$ . Further, assume that there is at least one point  $x_0 \in (a, b)$  s.t.  $\{f_n(x_0)\}$  converges. Then,

- There is a function  $f$  s.t.  $f_n \rightarrow f$  uniformly on  $(a, b)$ , and
- For each point in  $x$  in  $(a, b)$ , the derivative  $f'(x)$  exists and equals  $g(x)$ .

### Theorem 8.3.5

Upon its interval of convergence:

- The power series of a function converges uniformly to a continuous integrable
- The integral of the sum of the series is the sum of the integrals of the terms
- The derivatives of the power sums converge uniformly
- The partial sums of the series add to  $f(x_0)$  at  $x_0$
- Can differentiate a power series term-by-term to obtain the derivative of the function to which it converges.

## 9. Picard's theorem

### Definition 9.1 (Lipschitz in the second variable)

$F : \mathbb{R}^2 \rightarrow \mathbb{R}$  is **Lipschitz in the second variable** iff there exists  $L \in \mathbb{R}$  s.t.

$$|F(x, y) - F(x, z)| \leq L|y - z|$$

for all  $y, z \in J$  and  $x \in I$ .

### Theorem 9.2 (Picard's theorem)

Let  $I, J \in \mathbb{R}$  be closed bounded intervals with  $(x_0, y_0) \in I^\circ \times J^\circ$ .

If  $F : I \times J \rightarrow \mathbb{R}$  is continuous and Lipschitz in the second variable, then there exists  $h > 0$  s.t.  $[x_0 - h, x_0 + h] \subset I$  and a unique differentiable function  $f : [x_0 - h, x_0 + h] \rightarrow J$  s.t.

$$f'(x) = F(x, f(x)) \quad f(x_0) = y_0.$$

**In short**, if  $F$  is continuous and Lipschitz in the second variable, the initial value problem has a unique solution on some interval around  $x_0$ .

The unique solution can be constructed via the following procedure:

1. Define  $f_0(x) := y_0$ .
2. For each  $k \in \mathbb{N}^+$ , define the ***k*th Picard iterate**:

$$f_k(x) := y_0 + \int_{x_0}^x F(t, f_{k-1}(t)) dt$$

3. The sequence of Picard iterates converges uniformly to the solution  $f$ .

## 10. Lebesgue integrability condition

### 10.1. Oscillation

#### Definition 10.1.1 (Oscillation)

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  and  $S \subset \mathbb{R}$ . The **oscillation** of  $f$  on  $S$  is defined as

$$\omega_f(S) := \sup\{f(S)\} - \inf\{f(S)\}.$$

If the supremum or infimum does not exist, it is defined as  $+\infty$ .

The **oscillation** of  $f$  at the point  $x \in S$  is defined as

$$\lim_{\varepsilon \rightarrow 0} \omega_f(S \cap (x - \varepsilon, x + \varepsilon)).$$

### Theorem 10.1.2

$f$  is continuous at  $x_0$  iff the oscillation of  $f$  at  $x_0$  is zero.

## 10.2. Measure

### Definition 10.2.1 (Measure)

A **measure** is a function that takes a subset  $S$  of the reals, and returns its “size” denoted  $|S|$ , following the following properties:

- $0 \leq |S| \leq \infty$ .
- $|\emptyset| = 0$ .
- $S' := \{x + a : x \in S\} \implies |S'| = |S|$ .
- $|(0, 1)| = |[0, 1]| = 1$ .
- **Countably additive**: If  $\{S_i\}$  is a countable collection of disjoint sets,  $|\cup S_i| = \sum_i |S_i|$ .

### 10.2.1. Lebesgue outer measure

#### Definition 10.2.2 (Lebesgue outer measure)

Cover the set with a countable number of disjoint open intervals, and sum their totals. Take the infimum of all such coverings.

$$\lambda^*(S) := \inf\left\{\sum (b_i - a_i) \mid \{(a_i, b_i)\}_{i \in \mathbb{Z}^+} \text{ is such a cover}\right\}.$$

### Theorem 10.2.3

- Every countable subset of  $\mathbb{R}$  has [Lebesgue outer] measure 0.
- A subset of a measure 0 set also has measure 0.
- Any countable union of measure 0 sets has measure 0.

## 10.3. Lebesgue integrability condition

### Theorem 10.3.1 (Lebesgue Integrability Condition)

$f : [a, b] \rightarrow \mathbb{R}$  is Riemann-integrable iff it is bounded and its set of discontinuities has measure 0.

#### Corollary 10.3.1.1

Let  $f : [a, b] \rightarrow [c, d]$  be integrable and  $g : [c, d] \rightarrow \mathbb{R}$  continuous. Then,  $g \circ f$  is integrable.

## 11. Counterexamples & Constructions

### 11.1. Pathological Functions

**P1. Dirichlet function:**  $D(x) = 1$  if  $x \in \mathbb{Q}$ , 0 if  $x \notin \mathbb{Q}$ . *Properties:* bounded; nowhere continuous (every interval contains both rationals and irrationals); NOT integrable ( $U(P, D) = 1$ ,  $L(P, D) = 0$  for any partition  $P$ ); can be written as  $D(x) = \lim_{m \rightarrow \infty} (\lim_{n \rightarrow \infty} \cos^{2n}(m! \pi x))$ .

**P2. Thomae's function** (Modified Dirichlet):  $f(\frac{p}{q}) = \frac{1}{q}$  (lowest terms),  $f(\text{irrational}) = 0$ . *Properties:* bounded; continuous at every irrational (hence almost everywhere); discontinuous at every rational; Riemann integrable with  $\int_0^1 f = 0$ ; discontinuities are countable (measure 0). Shows the Lebesgue Integrability Condition in action.

**P3.**  $f(x) = x \sin(\frac{1}{x})$ ,  $f(0) = 0$ . *Properties:* continuous everywhere; uniformly continuous on  $[0, 1]$ ; NOT Lipschitz at 0 (oscillates arbitrarily steeply); NOT differentiable at 0.

**P4.**  $f(x) = x^2 \sin(\frac{1}{x})$ ,  $f(0) = 0$ . *Properties:* differentiable everywhere;  $f'(0) = 0$ ;  $f'(x) = 2x \sin(\frac{1}{x}) - \cos(\frac{1}{x})$  for  $x \neq 0$ ;  $f' \nrightarrow 0$  as  $x \rightarrow 0$ , so  $f'$  is NOT continuous at 0 ( $f$  is differentiable but  $f' \notin C^1$ ).

**P5.**  $f_{n(x)} = x^n$  on  $[0, 1]$ . Pointwise limit:  $f(x) = 0$  for  $x < 1$ ,  $f(1) = 1$  (discontinuous). Convergence is NOT uniform:  $\|f_n - f\|_\infty = 1$  for all  $n$ . Shows the uniform limit of continuous functions need not hold under pointwise convergence.

**P6. Weierstraß function:**  $W(x) = \sum_{n=0}^{\infty} (\frac{1}{2})^n \cos(3^n \pi x)$ . Continuous everywhere, nowhere differentiable ( $\|f_n - f\|_\infty \rightarrow 0$  by the Weierstraß  $M$ -test, but derivatives blow up). Limit of smooth functions can fail to be smooth.

**P7.**  $f(x) = \sqrt{x}$  on  $[0, 1]$ . Uniformly continuous (compact domain and continuous). NOT Lipschitz at 0:  $\frac{|\sqrt{x}-0|}{|x-0|} = \frac{1}{\sqrt{x}} \rightarrow \infty$ . Shows uniform continuity  $\nRightarrow$  Lipschitz.

**P8.**  $f(x) = \frac{1}{x}$  on  $(0, 1)$ . Continuous. NOT uniformly continuous ( $\delta$  must depend on position; Cauchy sequences map to non-Cauchy sequences). NOT bounded. Shows  $f : (a, b) \rightarrow \mathbb{R}$  is uniformly continuous if and only if it extends continuously to  $[a, b]$ .

**P9.**  $f(x) = |x|$ . Uniformly continuous; Lipschitz (constant  $K = 1$ ); NOT differentiable at 0. Shows Lipschitz  $\nRightarrow$  differentiable.

**P10.** If  $|f(x) - f(y)| \leq (x - y)^2$  for all  $x, y$ , then  $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = 0$  everywhere, so  $f$  is constant.

**P11.** The alternating harmonic series  $\sum \frac{(-1)^n}{n}$  converges (by the alternating series test) but NOT absolutely (since  $\sum \frac{1}{n}$  diverges). Any real number can be achieved by rearrangement (Riemann Rearrangement Theorem).

### 11.2. Property Constructions (No Proof)

• **Continuous, not uniformly continuous:**  $f(x) = x^2$  on  $\mathbb{R}$ ; or  $\sin(\frac{1}{x})$  on  $(0, 1]$ .

• **Uniformly continuous, not Lipschitz:**  $f(x) = \sqrt{x}$  on  $[0, 1]$ .

• **Lipschitz, not differentiable:**  $f(x) = |x|$  (constant  $K = 1$ ).

• **Pointwise convergent, not uniformly convergent:**  $f_{n(x)} = x^n$  on  $[0, 1]$ .

• **Uniformly convergent, but derivatives may not converge:**  $f_{n(x)} = \frac{\sin(nx)}{n} \rightarrow 0$  uniformly, but  $f_{n'}(x) = \cos(nx) \nrightarrow 0$ .

• **Bounded, not integrable:** Dirichlet function  $D(x)$ .

• **Integrable, not continuous (even with dense discontinuities):** Thomae's function.

• **Integrable  $f$ , injective  $g$ , but  $f \circ g$  not integrable:** Take  $f = \text{Thomae}$  (discontinuous on  $\mathbb{Q}$ ) and  $g$  a bijection mapping irrationals to rationals (e.g. Cantor-based);  $f \circ g$  is 0 on irrationals,  $\frac{1}{q}$  on images of rationals — can be made non-integrable.

• **Sequence with  $\limsup \neq \liminf$ :**  $\{(-1)^n\}$ :  $\limsup = 1$ ,  $\liminf = -1$ .

• **Limit comparison test failure:**  $\sum \frac{1}{n}$  and  $\sum \frac{1}{n^2}$ : ratio  $\rightarrow \infty$ , both diverge/converge differently — shows the limit comparison test requires  $0 < L < \infty$ .

### 11.3. Quick Reference: Implications

- Lipschitz  $\Rightarrow$  Uniformly continuous  $\Rightarrow$  Continuous
- Differentiable  $\Rightarrow$  Continuous
- Uniformly convergent  $\Rightarrow$  Pointwise convergent
- Absolutely convergent  $\Rightarrow$  Convergent
- Closed + bounded  $\Leftrightarrow$  Sequentially compact
- Continuous  $\Rightarrow$  Integrable

None of these reverse in general.